

Research Note — Disrupt Quant 2026

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1. Hypothesis

I initially tried to find whether an asset's recent price momentum or reversal would predict near-term future returns. I tested this two ways. First, using how much the asset had moved over the past 20 days to predict what it would do over the next 5 days, and second, checking whether an asset was beating or lagging its sector on a given day predicted what happened next. I picked these because momentum and reversal are well known patterns in real financial markets, so I believed it would make a great starting point.

2. Evidence

For both hypotheses, I split the data into 5 groups, from "moved down the most" to "moved up the most," and checked the average future return in each group. If there were a real pattern, the groups should have lined up in order, but they didn't. Neither test showed a clean, monotonic relationship, as return_20d buckets showed values ranging narrowly from about 0.0002 to 0.0014 with no consistent increasing or decreasing order, in addition to the sector-relative test showing a similarly small, inconsistent pattern. A pooled scatter plot of return_20d against forward returns also showed a roughly symmetric cloud with no visible directional relationship. I concluded that neither signal was strong or consistent enough to trust with real money, and decided not to keep testing more columns, since testing too many things increases the chance of "finding" a fake pattern.

3. Construction

Given the absence of a reliable directional signal, I decided not to predict returns. Instead, I built a strategy that focuses on an asset's recent volatility, instead of which direction it's moving. This gives calmer assets moderately more capital and volatile assets moderately less, as a risk-management choice rather than a return-prediction claim. Additionally, I used a softened version of this formula (an inverse square root instead of a plain inverse) after testing both and finding the softer version had lower trading costs with almost the same results.

4. Risk

The strategy carries no leverage and no short positions, as my net exposure equaled 100%. This allows my total risk to stay capped at exactly the amount of money that I have. I spread the money across 24 assets and 6 sectors, limiting the amount of damage from a single asset. However, an issue I'm aware of is the fact that my strategy measures the magnitude of an asset's price movements, not their direction. That being said, my strategy cannot distinguish a stable, slowly-declining asset from a stable, flat or

rising one, and would allocate more capital to a "calm decliner" than a jumpy outperformer.

5. Costs

Because target weights change only when volatility estimates shift, turnover was low, with 2.3% of NAV/day on development, and 3.7%/day on validation. Cumulative transaction costs were small relative to NAV (1.0% over the full development period, 0.24% over validation). This directly reflects the design intent, as with no trading signal to justify frequent rebalancing, minimizing unnecessary turnover was a priority.

6. Validation

I tested my strategy on the two separate chunks of data. The first was the development period, which spanned from Jan 2021-Jun 2024, producing about 14.3% in total return, annualized 3.8%, Sharpe 0.38, and max drawdown -10.8% in 910 sessions.

Additionally, the validation period showed loss. I do not view the validation period as strong evidence, as my strategy was not designed to predict direction, meaning losing a 6-month window is not a flaw in design logic, although it was a disappointing result.

After seeing that results, I tried a stronger version of the same idea just to see if performance would improve, but instead produced something almost identical.

Therefore, to limit trading costs, I stuck with the improved version.

7. Limitations

With only 132 validation sessions, it is difficult to distinguish "this approach doesn't help" from "one unlucky 6-month stretch." The strategy provides no protection against steady directional decline in low-volatility assets. I did not test robustness across sectors or sub-periods within development, so the (already weak) momentum/reversal findings and the volatility-tilt's benefit are not confirmed to be stable across sub-periods. I also did not run a formal equal-weight baseline comparison, because by construction, this strategy differs from equal-weight only in reallocating weight toward lower-volatility assets.

8. Next step

The most useful follow up test would be to split money evenly across every sector to determine if the volatility-based weighting is actually adding value, or just introducing complexity for no real benefit. I would also re-check my momentum/reversal tests separately by sector and by year, rather than all mixing them together, since a real pattern could be hiding in a subset of the data that got averaged away in my first pass.